

Software Requirements Specification

Mboa Compiler Construction Lab

Real manual statements. Deterministic lexical and syntactic analysis.

NON-AI / PRIVATE / EVIDENCE-LED

This document specifies the assignment-focused software, not the final coursework report. Genuine fieldwork, an evidence-based grammar, real screenshots, linguistic discussion and final pagination still require the group.

AUTHOR GROUP

Kwete Ngouba Junior Rayan	ICTU20241377
Djemtchimo Noukui Bruno Jonatan	ICTU20241585
Amina Boubakary	ICTU20241870

Version 3.0 | 22 September 2026

Assignment due: 29 September 2026

Contents

1	Purpose and Document Control	2
1.1	Status vocabulary	2
2	Assignment Baseline and Traceability	2
3	Product Scope and Actors	3
3.1	Primary roles	3
3.2	Included and excluded capabilities	4
4	Functional Requirements	4
5	Quality Requirements and Failure Contracts	6
6	Data, Limits and Lifecycle	7
6.1	Data classifications	7
6.2	Bounded operation	8
6.3	Backup compatibility	8
7	Interfaces and Operating Constraints	9
8	Acceptance Strategy and Completion Gates	9
8.1	Software verification	9
8.2	Fieldwork and report gates	9

1 Purpose and Document Control

Mboa supports the CS4110 Compiler Construction SET A assignment, *Lexical and Syntactic Analysis of Informal Urban Communication in Yaoundé*. The product combines a deterministic custom lexer and LL(1) compiler laboratory with private manual collection, local text import and recoverable account/project storage. It is not an AI translation or conversational service.

Version 3.0 supersedes the active business/AI product scope. Earlier `*.legacy.md`, `srs-legacy.tex`, `sdd-legacy.tex` and evidence JSON remain historical material; they are not acceptance of this release. No earlier research or sprint is retroactively declared complete. The SDD and its source-grounded UML explain implementation. The root README explains operation; the documentation register records fresh document validation.

1.1 Status vocabulary

Implemented capability

A source-level function or user workflow exists. This alone does not prove correct fieldwork, complete coursework or every deployment configuration.

Automated evidence

A stated test/build/measurement with controlled fixtures. Synthetic fixtures never count as real collected statements.

Human gate

Authentic transcription, linguistic interpretation, consent, final report review, rehearsal or deployment acceptance.

Illustrative material

Supplied references, constructed examples and the default CFG. These are learning aids, not empirical findings.

Current evidence boundary

At the 22 September 2026 scope review, the legacy CSV and three inspected hosted workspaces each contained zero entries. The user's genuine manually transcribed text file is pending. The 179 reference rows and 26 constructed examples are not fieldwork; the default CFG is not corpus-derived.

2 Assignment Baseline and Traceability

The supplied Summer 2026 brief sets the due date to **29 September 2026**. Groups contain three students. The final report must be 25–30 pages and must not exceed 30 pages. A ten-minute PowerPoint presentation and demonstration allocates three minutes to each member, leaving one shared minute.

Suggested contexts include taxis/commuting, internet connectivity, electricity, market bargaining, rain, fuel scarcity, roadside business, bendskins, security and university life. The brief does not require a fabricated example for every topic.

Brief requirement	Software support	Required human evidence
10–15 real statements; group of three	Sentence counts, private collection, provenance fields, three-member profile, manual-transcription self-attestation	Real listening/recording and faithful manual transcription; no generated or constructed replacements
Nouns, verbs, slang, code-mixing and multiword phrases	Regex/lexicon specification, token tables, phrase detection and unknown labels	Review classification and explain multilingual ambiguities
Custom lexer; frequency and variation	Python regex implementation; corpus token/category counts; spelling variants	Interpret observed counts and differences, not hypothetical findings
Construct CFG; remove left recursion; left-factor	Editable CFG and ordered transformation logs with before/after productions	Tie rules and rationale to structures in the actual statements
FIRST/FOLLOW and LL(1) table, or LR alternative	Fixed-point FIRST/FOLLOW; predictive table and conflicts; LL(1) parser	Explain calculated artifacts and any unresolved conflicts. LR/SLR is not implemented or required in addition to the chosen LL(1) path
Parse own tokenized inputs; accept/reject	Single-input trace and full saved-corpus outcomes	Own-data test cases and a discussion of rejections/limitations
25–30-page report	Draft HTML structure, raw/tokens/regex/grammar/table artifacts and real screenshot attachments	Original linguistic discussion, screenshot authenticity and actual page count review, capped at 30
Source and ten-minute demo	Source snapshots, own-corpus cases and editable slide draft with timing notes	Review source/cases, personalize slides and rehearse three minutes per member

3 Product Scope and Actors

3.1 Primary roles

A visitor can register, verify email, sign in and recover an account. A verified account owner works only inside that account’s private projects. Three group names are assignment metadata, not a shared-team authorization role. There is no cross-account collaboration, billing or in-app administrator console.

The operator manages the durable host, identity/email configuration and backup policy. The operator can access underlying files; account isolation is not end-to-end encryption. SMTP delivers account messages in production. Optional Google OAuth supplies identity, never language generation.

3.2 Included and excluded capabilities

The default authenticated application exposes compiler/coursework workflows, manual collection, reference dictionary, constructed practice, local document import, retained history and workspace settings. Accounts, projects, recordings, revisions and backups remain available.

The product excludes AI generation, chat, translation endpoints, provider explanations, OCR, remote transcription and browser dictation. Leftover private GEMINI_* configuration is unused; no model key is required. Optional browser text-to-speech is read-aloud only, not field transcription or linguistic evidence. Only voices reported by the browser as `localService` are selected, with no remote-voice fallback. Voice availability and locality reporting depend on the browser/OS; this is not an application-level guarantee of every underlying network behavior.

The desktop collector and an explicitly unauthenticated maintenance factory retain compatibility with legacy files. They are separate from normal web storage. Production refuses to disable authentication.

4 Functional Requirements

ID / area	Requirement and measurable acceptance
FR-01 Accounts	Require verified email/password access; support explicit optional Google sign-in/linking and single-use verification/recovery tokens. Acceptance: Unverified password login fails. Invalid/replayed tokens fail. Recovery revokes previous sessions. Matching Google email alone cannot link an account.
FR-02 Sessions	Enforce authenticated private APIs, allowed Origin and session-specific CSRF on mutations. Keep deployment credentials server-side. Acceptance: Missing identity returns 401; invalid Origin/CSRF returns 403. Production requires secure cookies/HTTPS and refuses no-auth operation.
FR-03 Isolation	Select the account directory from server-validated identity and bind dataset and coursework adapters to the current project using request-local contexts. Acceptance: Foreign account/project access cannot expose or mutate statements, profile, screenshots, audio, history, revisions or backups. Concurrent requests do not switch one another's storage.
FR-04 Projects	Create/rename projects, retain General, and allow deletion only when the project is empty of protected records, including coursework/screenshots. Acceptance: An unknown/foreign project fails. Switching/sign-out warns before discarding unsaved drafts. Existing accounts are neither deleted nor silently migrated.
FR-05 Manual data	Preserve exact nonblank raw statements, glosses and notes, including case, accents, code-mixing and incomplete speech. Store provenance separately. Acceptance: Raw text, glosses and notes are not rewritten. Existing ShortText validation trims supplied contributor/location values on save, not during import preview. Duplicates and failed saves are explicit. Only Sentence entries count toward the 10–15 requirement.

ID / area	Requirement and measurable acceptance
FR-06 Review	Require explicit human review of annotations. An evidence edit invalidates prior approval unless the new evidence is explicitly reviewed. Acceptance: Imported records start unreviewed. A change of raw text, language or lexical annotation cannot silently keep stale approval.
FR-07 Recording	Allow local browser recording/preview/download and explicit private attachment, without dictation, transcription or automatic upload on recording alone. Acceptance: Permission/codecs failures are actionable; typed/file input remains available. Private playback honors owner/project selection and HTTP range semantics.
FR-08 Import	Extract UTF-8 TXT/Markdown/CSV/JSON, DOCX and text-layer PDF locally on the application server. Show text/warnings/drafts before any save. Acceptance: No generation/provider call occurs. Unsupported/scanned-only inputs fail explicitly. Structured metadata cannot import an approved status. Extracted text is bounded and checked against the original by the user. Preview preserves provenance values; an explicit save applies the short-metadata rule in FR-05.
FR-09 References	Expose a read-only, source-labelled dictionary and clearly constructed practice library without populating the real corpus automatically. Acceptance: Search preserves reference file/line and competing senses. Reference origin is not presented as POS or a French translation; practice is marked constructed.
FR-10 Lexer	Use a deterministic custom regex/lexicon pipeline for token categories, multiword verb/slang phrases and code-mixing clues, preserving raw token text. Acceptance: Token tables retain unrecognized symbols as UNKNOWN. Approved unambiguous Word annotations can extend lookup; Sentence rows do not train a model.
FR-11 Corpus metrics	Compute frequency, category totals, unknown tokens and observed spelling variation from the selected project's saved entries. Acceptance: No reference/example rows are injected. Normalized variation groups are labelled orthographic candidates, not certified semantic equivalence.
FR-12 Project profile	Save exactly three member strings, manual-transcription confirmation, collection method, grammar, rationale, discussion and limitations per project. Acceptance: Nonblank member names must be distinct. New profiles use blank members and an illustrative grammar; missing fieldwork never becomes an automatic ready claim.
FR-13 Grammar	Parse bounded CFG notation, remove supported direct/indirect left recursion and left-factor with traceable before/after steps. Acceptance: Invalid notation or excessive transformation work is rejected. Unchanged stages are explained. The group must justify the grammar against its own data.

ID / area	Requirement and measurable acceptance
FR-14 Sets/table	Compute FIRST/FOLLOW to a fixed point, including epsilon/end marker, then construct an LL(1) table with explicit conflicts. Acceptance: Conflicts and residual recursion are surfaced; the implementation does not arbitrarily select a production to conceal a non-LL(1) grammar.
FR-15 Parser	Run a bounded table-driven predictive parser over lexical categories, showing stack, remaining input, selected production/action and acceptance/rejection. Acceptance: Single-input and corpus results use the submitted grammar and actual tokenized text. Rejection is a grammar-fit result, not a judgment of spoken correctness.
FR-16 Screenshots	Accept genuine uploaded analyzer captures with bounded decoding, normalization and private owner/project-scoped byte retrieval. Acceptance: Only allowed image formats/sizes pass. Foreign project/owner reads fail. Upload/delete increments the workspace version; no fabricated evidence is generated.
FR-17 Course export	Export a DRAFT HTML report, editable PowerPoint, saved raw data, lexical/grammar/ parser artifacts, screenshot bytes and implementation source snapshots. Hold the account's shared lock across the profile/corpus/screenshot evidence read. Acceptance: Missing data/discussion is explicit. The 25-section structure is not called 25 printed pages. Concurrent account mutations cannot mix saved evidence revisions during export. Report/presentation remain human-review tasks.
FR-18 Legacy history	Keep existing AI-generated translations/conversations readable and exportable as legacy data without resuming a generation workflow. Acceptance: Upgrading the product does not delete histories or invoke removed endpoints when viewing/downloading them.
FR-19 Recovery	Retain revisions, verified backups, bounded scheduling, validation previews, version checks and a pre-restore safety backup. Include coursework/screenshots. Acceptance: Workspace document format 2 round-trips private data; legacy format 1 remains restorable. The outer archive manifest stays format 1. Stale previews cannot overwrite new profile/screenshot changes.
FR-20 Non-AI contract	Return health status ok with mode compiler. Remove translation/chat/suggestion/ explanation routes and generation settings/quotas. Acceptance: No language-provider client or model configuration is required. Local compiler and import tests run without external providers; Google HTTP and SMTP remain separate identity facilities.

5 Quality Requirements and Failure Contracts

ID	Requirement and evidence boundary
QR-01 Integrity	Use SQLite transactions and coordinated file locks. Reject stale recovery versions, invalid identifiers, unsafe archive paths and invalid media. Filesystem and SQLite operations are coordinated, not a distributed ACID transaction. Failures must not be reported as successful persistence.
QR-02 Privacy	No field data or source code is sent to a generation provider or online diagram renderer. Private account storage remains outside public assets. Protect session/ email/OAuth secrets and avoid query-string logging. The server/operator remains inside the trust boundary.
QR-03 Explainability	Expose rules, transformation steps, sets, table conflicts, traces and lexical limitations. Natural-language explanations are authored by students, not generated. A passing parse is not evidence of real-world authenticity.
QR-04 Responsiveness	Bound file, corpus, grammar and trace work; preserve cancellation/stale-response protection in the UI. Report measured timings with machine, fixture, warm-up, sample count and limitations, never borrowed historical performance claims.
QR-05 Accessibility	Retain labelled forms, keyboard actions, focus-managed dialogs, visible errors and text/file alternatives to microphone functions. Test desktop/mobile layouts; mocked checks alone do not certify every browser, device or assistive technology.
QR-06 Maintainability	Use existing test/build tools. Cover all authored production classes in UML with actual methods/relationships. Validate every sequence activation, atlas mapping, image fingerprint and clean LaTeX references/layout after rebuilding.

Validation errors use actionable messages and bounded inputs. Authentication, authorization, conflict, missing-resource and storage failures remain distinct; provider availability is not a compiler failure category. A failed PDF import must request a supported/manual transcript rather than silently fabricate text.

6 Data, Limits and Lifecycle

6.1 Data classifications

Data	Purpose and storage	Evidence status
Raw state-ments/provenance	Selected-project entries and revisions in the account SQLite database; original wording separate from glosses	Human-supplied, authenticity not machine-proven
Coursework profile	Project-keyed JSON with three names, grammar and narrative	Human-authored metadata, not a generated report
Screenshots	Project-bound ID/name/base64 PNG rows with private byte route	User must provide genuine captures

Data	Purpose and storage	Evidence status
Recordings	Private account audio files referenced by entries	Supporting raw material, never auto-transcribed
Dictionary and practice	Versioned source-labelled references and constructed examples	Not collected statements
Legacy history	Existing validated translation/conversation records	Historical output, no new generation
Backups	Verified account ZIPs containing all project data/media	Sensitive recovery copies, not proof of authorship

6.2 Bounded operation

Boundary	Limit
Raw entry / individual analysis	4,000 characters
Coursework corpus analysis	500 entries; 100,000 text characters
Grammar text	12,000 characters plus syntax/transformation/parser work guards
Import input / extracted text	12 MiB; 40,000 characters
PDF / structured drafts	40 pages; 100 draft records
Screenshots per selected project	6; 2 MiB each; 12 megapixels
Projects / entries per account	20; 10,000
Saved history / revisions per account	500; 20,000
Saved conversation messages	1–100, for legacy history compatibility
Private recordings	1,000 files; 128 MiB aggregate
Backup ZIP compressed / expanded	32 MiB; 160 MiB
Stored backups / pending previews	512 MiB per account; three one-hour previews
Automatic backup retention	1–10 copies, daily or weekly

These are storage/processing limits, not assignment targets or certified load capacity. Deletion may leave older revisions/backups until their explicit retention policy removes them. Unreferenced recordings are not silently erased. The screenshot route allows a 3 MiB JSON body for base64 transport; the generic JSON cap remains 1 MiB. Coursework exports contain screenshot bytes but do not bundle raw recordings; full workspace backups preserve recordings and coursework together.

6.3 Backup compatibility

The workspace document advances to format 2 to include coursework and screenshots. Restoring format 1 remains supported without those newer collections. The outer ZIP manifest remains format 1. Every new coursework/screenshot mutation must advance the workspace version so a preview cannot overwrite unseen changes. Restore validates logical records, stages safe media, creates a safety backup and commits under the selected account lock.

Backups omit authentication secrets and server configuration. The operator must protect and rehearse a separate consistent backup of the full data root, including the identity database. Same-disk workspace ZIPs cannot recover a lost host or recreate account identities on their own.

7 Interfaces and Operating Constraints

The browser uses same-origin HTTP APIs served by FastAPI alongside the built React application. Private calls carry a session cookie; mutations additionally require Origin/CSRF. A validated project header selects request scope; private native audio/image URLs can carry a project query parameter.

Area	Interface
Compiler	<code>/api/analyze</code> ; coursework state, project, analyze, parse, screenshots and export under <code>/api/coursework</code>
Collection and references	Dataset CRUD/audio, metadata, dictionary and constructed examples under <code>/api</code>
Import	<code>/api/imports/preview</code> ; supported local text formats only
Accounts/recovery	Auth/session/Google routes and private project, legacy history, revision and backup routes
Health	<code>status=ok, mode=compiler</code> ; not provider acceptance
Removed interfaces	<code>/api/translate</code> , <code>/api/chat</code> , <code>/api/imports/suggest</code> , <code>/api/coursework/explain</code>

Local operation uses loopback and a private file-mail outbox unless SMTP is configured. Production requires an HTTPS public origin, secure cookies, actual SMTP and durable private storage behind a trusted reverse proxy; interactive API docs are disabled. Google remains optional. The supported topology is one durable application host, not a stateless/multi-host SQLite cluster.

8 Acceptance Strategy and Completion Gates

8.1 Software verification

Fresh backend/compiler, frontend and browser results must name exact commands, fixtures and outcomes. Unit tests use synthetic data and isolated storage; browser tests exercise the built application. Compiler measurements must identify the measured workload and exclusions. The documentation build must render all current diagrams, compile atlas/SRS/SDD, and pass the strict checker without layout/reference warnings. Earlier evidence JSON stays historical and unmodified.

8.2 Fieldwork and report gates

1. Receive 10–15 genuine manual transcriptions and record provenance, collection method, consent/anonymization and all three member contributions.
2. Review exact raw text and lexical labels, including unknowns, multiword phrases, code-mixing and frequency/variation interpretations.
3. Construct and justify the CFG from those observations; examine transformation logs, FIRST/FOLLOW, conflicts and own-data accepted/rejected cases.
4. Attach genuine analyzer screenshots and author the linguistic discussion: mixing, slang, context, incomplete speech and limits of the chosen formal grammar.
5. Export the draft and check actual printed pagination: 25–30 pages, never over 30. The engineering SRS/SDD are not that report.

6. Rehearse the ten-minute demo with three minutes per member and a shared minute. Only the group/instructor can approve submission readiness.

No premature completion claim

Empty data, an illustrative CFG, synthetic test passes or a generated draft cannot satisfy real collection and linguistic analysis. The product deliberately keeps authenticity, report pagination and presentation rehearsal as human gates.